MATHEMATICS AND STATISTICS

Time: 3 Hrs. **Max. Marks: 80**

General instructions:

*The question paper is divided into **FOUR** sections.*

(1) ***Section A:*** *Q. 1 contains **Eight** multiple choice type questions, each carrying **Two** marks. Q. 2 contains **Four** very short answer type questions, each carrying **One** mark.*

(2) ***Section B:*** *Q.3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)*

(3) ***Section C:*** *Q.15 to Q.26 contains **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)*

(4) ***Section D:*** *Q.27 to Q.34 contains **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)*

(5) Use of log table is allowed. Use of calculator is not allowed.

(6) Figures to the right indicate full marks.

(7) Use of graph paper is not necessary. Only rough sketch of graph is expected.

(8) For each multiple choice type of questions, only the first attempt will be considered for evaluation.

(9) Start answer to each section on a new page.

SECTION – A

Q.1. Select and write the correct answer for the following multiple choice type of questions: **[16]**

i. If A = {1, 2, 3, 4, 5} then which of the following is not true? (2)

(A) $\exists\, x \in A$ such that $x + 3 = 8$ (B) $\exists\, x \in A$ such that $x + 2 < 9$

(C) $\forall\, x \in A, x + 6 \geq 9$ (D) $\forall\, x \in A, x + 5 \leq 10$

ii. The value of $\sin\left[\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) - 3\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right] =$ (2)

(A) $-\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $-\frac{\sqrt{3}}{2}$

iii. If $\hat{p}, \hat{q}, \hat{r}$ are mutually perpendicular unit vectors and form a right handed triplet then the value of $\hat{p}\cdot(\hat{q} \times \hat{r}) + \hat{q}\cdot(\hat{p} \times \hat{r}) + \hat{r}\cdot(\hat{p} \times \hat{q}) =$..... (2)

(A) –2 (B) 0 (C) 1 (D) 3

iv. The Cartesian equation of the line passing through the points A(3, 4, –1) and B(2, –1, 3) is _____. (2)

(A) $\frac{x+3}{1} = \frac{y-4}{5} = \frac{z+1}{-4}$ (B) $\frac{x-3}{-1} = \frac{y-4}{-5} = \frac{z+1}{-4}$

(C) $\frac{x-3}{-1} = \frac{y-4}{-5} = \frac{z+1}{4}$ (D) $\frac{x-3}{-1} = \frac{y-4}{5} = \frac{z+1}{-4}$

v. If $y = \log_x a$ then find $\frac{dy}{dx} =$ _____. (2)

(A) $\frac{-\log a}{x(\log x)^2}$ (B) $x \log a$ (C) $-\frac{1}{\log_x a}$ (D) $\frac{1}{x \log a}$

vi. If $f(x) = \frac{\sin^{-1}x}{\sqrt{1-x^2}}$ and $g(x) = e^{\sin^{-1}x}$ then

$\int f(x)\cdot g(x)\, dx =$ _____. (2)

(A) $e^{\sin^{-1}x}(1 - \sin^{-1}x) + c$ (B) $e^{\sin^{-1}x}(\sin^{-1}x - 1) + c$

(C) $e^{\sin^{-1}x}(\sin^{-1}x + 1) + c$ (D) $e^{\cos^{-1}x}(\cos^{-1}x - 1) + c$

vii. The area bounded by the line $y = 2x$, X – axis and the lines $x = -1, x = 4$ is _____sq. units (2)

(A) 15 (B) 17 (C) 16 (D) 14

viii. In a Binomial Distribution, E(X) = 6, Var(X) = 4.2 then the value of n is _____. (2)

(A) 22 (B) 24 (C) 20 (D) 26

Q.2. Answer the following questions: **[4]**

i. Write the dual of $p \wedge t$. (1)

ii. What is the distance of point (3, –5, 6) from XZ plane? (1)

iii. The displacement of a particle at time t is given by $S = 3t^2 + 4t + 1$. Find its velocity at time t. (1)

iv. Write the order of differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{5}{2}} = \sqrt[3]{\frac{dy}{dx}}$ (1)

SECTION – B

Attempt any EIGHT of the following questions: **[16]**

Q.3. Construct the switching circuit for the statement $(\sim p \wedge q) \vee (p \wedge \sim r)$. (2)

Q.4. Find the matrix of cofactors of matrix $\begin{bmatrix} 5 & 2 \\ -1 & 6 \end{bmatrix}$. (2)

Q.5. Find the general solution of equation $\cos 5\theta = \frac{1}{2}$ (2)

Q.6. Find the value of k, if $2x + y = 0$ is one of the lines represented by $3x^2 + kxy + 2y^2 = 0$. (2)

Q.7. If $\bar{a}, \bar{b}, \bar{c}$ are the position vectors of points A, B, C respectively and $10\bar{a} = 7\bar{b} + 3\bar{c}$ then find the ratio in which the point C divides the line segment AB. (2)

Q.8. Find the angle between the lines

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$

$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(\hat{i} + 2\hat{j} + \hat{k})$ (2)

Q.9. Test whether the function $f(x) = x - \frac{1}{x}$, $x \in R$, $x \neq 0$ is increasing or decreasing. (2)

Q.10. Evaluate: $\int_{-1}^{1} |x|\, dx$ (2)

Q.11. Evaluate: $\int x \log x\, dx$ (2)

Q.12. Find the area bounded by curve $x^2 + y = 0$, X – axis and lines $x = 1, x = 4$. (2)

Q.13. The probability distribution of r.v. X is as follows:

$X = x$	0	1	2	3	4
$P(X = x)$	0.1	k	2k	2k	k

Find: (i) k (ii) F(2) (2)

Q.14. Evaluate: $\int \frac{e^x(1+x)}{\cos^2(xe^x)}\, dx$ (2)

SECTION – C

Attempt any EIGHT of the following questions: **[24]**

Q.15. Using truth table, determine whether statement pattern$[(p \vee q) \wedge \sim p] \wedge \sim q$ is a tautology or a contradiction or contingency. (3)

Q.16. Prove that:

$\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \sin^{-1}\left(\frac{56}{65}\right)$ (3)

Q.17. In ΔABC, if a = 10, b = 8, c = 6 then find the value of

(i) $\sin\left(\frac{C}{2}\right)$ (ii) $\cos\left(\frac{B}{2}\right)$ (3)

Q.18. Are the four points A(1, 2, 1), B(2, –3, 4), C(3, 4, –5), D(2, 3, –2) coplanar? Justify your answer. (3)

Q.19. Find the distance between parallel lines.

$\frac{x}{2} = \frac{y}{-1} = \frac{z}{2}$ and

$\frac{x+1}{2} = \frac{y-1}{-1} = \frac{z+1}{2}$ (3)

Q.20. Find the vector equation of the plane passing through the points A(2, 1, 1), B(0, 2, 3), C(4, 5, 6). (3)

Q.21. If $y = \log\left[\frac{x+\sqrt{x^2+a^2}}{\sqrt{x^2+a^2}-x}\right]$, find $\frac{dy}{dx}$. (3)

Q.22. A wire of length 36 cm is bent to form a rectangle. Find its dimensions, if the area of the rectangle is maximum. (3)

Q.23. Verify Rolle's theorem for the function $f(x) = x^2 - 5x + 9$, $x \in [1, 4]$ (3)

Q.24. Solve the differential equation $x\frac{dy}{dx} - \sqrt{x^2+y^2} = y$ (3)

Q.25. The following is the p.d.f. of continuous r.v.

$f(x) = \frac{x}{8}$; $0 < x < 4$

$= 0$, Otherwise

Find (i) expression for c.d.f of X.

(ii) F(x) at 0.5 and 5 (3)

Q.26. If the p.m.f of r.v. X be

$P(X = x) = {}^4C_x \left(\frac{5}{9}\right)^x \left(\frac{4}{9}\right)^{4-x}$ for $x = 0, 1, 2, 3, 4$

$= 0$; otherwise

then find E(X) and Var (X). (3)

SECTION – D

Attempt any FIVE of the following questions: **[20]**

Q.27. Find the inverse of matrix $\begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ (4)

Q.28. Prove that the homogeneous equation of degree two in x and y, $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin if $h^2 - ab \geq 0$. (4)

Q.29. Prove that if $\bar{a}, \bar{b}, \bar{c}$ are three non coplanar vectors, then any vector $\bar{r}$ in the space can be uniquely expressed as a linear combination of $\bar{a}, \bar{b}, \bar{c}$. (4)

Q.30. Solve the linear programming problem (L.P.P.) by graphical method.

Minimize: $z = 8x + 10y$

Subject to, $2x + y \geq 7$

$2x + 3y \geq 15$

$x \geq 0$,

$y \geq 2$ (4)

Q.31. If $x = f(t)$ and $y = g(t)$ are differentiable functions of t, so that y is differentiable function of x and $\frac{dx}{dt} \neq 0$ then prove that $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.

Hence find $\frac{dy}{dx}$, if $x = \sin t$, $y = \cos t$. (4)

Q.32. Evaluate: $\int \frac{2}{3 + 2\sin^2 x + 5\cos^2 x}\, dx$ (4)

Q.33. Prove that,

$\int_{-a}^{a} f(x)\, dx = 2\int_0^a f(x)\, dx$, if $f(x)$is even

$= 0$, if $f(x)$is odd. (4)

Q.34. A body cools according to Newton's law from 100°C to 60°C in 20 minutes. The temperature of the surrounding being 20°C. How long will it take to cool to 40°C? (4)